

if the corresponding bit offset is set to true.

This scheme for encrypted messages effectively divides the message counter space in half: those counters that are forward of the max message counter, which are considered new, and those counters that are behind the max message counter, which are considered duplicates unless indicated otherwise by the values in the bitmap.

4.6.5.3. Use of Message Reception State for Unencrypted Messages

For unencrypted messages, the algorithms for tracking messages and detecting duplicates are similar to, but more permissive than for encrypted messages using [Section 4.6.5.2.2, “Message Counters with rollover”](#). This reflects the fact that duplicate detection of unencrypted messages is not done for security reasons, but rather to catch duplicates caused by network errors (e.g. loss of an ack), which are generally more bounded in time. The more relaxed algorithm for unencrypted duplicate detection also relaxes the durability requirement on the sender’s message counter, allowing senders to store the counter in RAM.

For unencrypted messages, any message counter equal to `max_message_counter` or within the message counter window, where the corresponding bit is set to true SHALL be considered duplicate. All other message counters, whether behind the window or ahead of `max_message_counter`, are considered new and SHALL update `max_message_counter` and shift the window accordingly. Messages with a counter behind the window are likely caused by a node rebooting and are thus processed as rolling back the window to the current location. Note that when `max_message_counter` is close to the minimum of the range, the window SHALL roll back to cover message counters near the maximum of the range.

4.6.5.4. Message Reception State Initialization

To initialize [Message Reception State](#) for a given Peer Node ID, initial `max_message_counter`, Message Type (control or data), Encryption Level (encrypted or unencrypted), and Encryption Key (for any Encryption Level except unencrypted):

- The Message Reception State fields SHALL be set as follows:
 - The Peer Node ID SHALL reference the given Peer Node ID.
 - The Message Type SHALL be the given Message Type.
 - The Encryption Level SHALL be the given Encryption Level.
 - If the Encryption Level is NOT unencrypted, the Encryption Key SHALL reference the given key.
 - The `max_message_counter` SHALL be set to the given `max_message_counter`.
 - The Message Counter bitmap SHALL be set to all 1, indicating that only new messages with counter greater than `max_message_counter` SHALL be accepted.

4.6.6. Counter Processing of Outgoing Messages

1. Obtain the outgoing message counter of the sending Node for the given Security Flags, Session Id, and Encryption Key: